Armita Hoda

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RESEARCH INTERESTS

Chromatin regulation, Neurodevelopmental disorders, (Epi)genome editing, RNA therapies

EDUCATION

University of Tehran, Tehran, Iran **2020-2026**
Department of Biotechnology (Integrated B.Sc.–M.Sc.–Ph.D. Program)
GPA **3.92**
M.S. Phase Thesis: *Epigenome-aware mechanistic modeling of cfDNA fragmentation dynamics*

Farzanegan School, Babol, Iran **2012-2019**
National Organization for Development of Exceptional Talents

RESEARCH EXPERIENCE

Cell Type Specific Partitioned Heritability of Psychiatric Disorders
Stratified LD Score Regression (S-LDSC) • *Independent Project* July 2026 – present

- Built a computational pipeline linking GWAS summary statistics of neurodevelopmental traits (OCD, Anorexia, ADHD, Schizophrenia, Tourette Syndrome) to chromatin accessibility data (human brain snATAC-seq)
- Computed partitioned LD scores and ran S-LDSC regression against both the baseline v1.2 and baselineLD v2.2 models to jointly assess cell-type significance and heritability enrichment magnitude
- Re-implemented the full workflow as a reproducible Snakemake pipeline for incremental extension to new cell types and traits
- Preliminary results: identified significant heritability enrichment in cortical excitatory neurons, striatal medium spiny neurons, and cortical interneuron subtypes, with markedly weaker enrichment in glial/non-neuronal controls

Epigenome-aware Mechanistic Modeling of Cell-free DNA Fragmentation **Tehran, Iran**
Master's Thesis • *Department of Biotechnology, University of Tehran, Iran*
June 2025 - present

- Designed a continuous-time Monte Carlo algorithm capable of tracking fragment birth, cleavage, immigration, and clearance while preserving fragment genomic coordinates throughout the simulation of cfDNA
- Integrated nucleosome positioning and chromatin accessibility into a genomic-coordinate-aware simulation of cfDNA fragmentation derived from MNase and ATAC-seq datasets
- Developed probabilistic cleavage kernels for DNase1, DNase1L3, and DFFB that combine sequence motifs with chromatin context to simulate biologically realistic fragmentation events

- Simulated experimentally measurable fragmentomic signals including fragment length spectra, fragment-end sequence motifs, Windowed Protection Scores (WPS), and nucleosome spacing distributions for direct comparison with sequencing data
- Validated simulated fragmentomic profiles against healthy plasma and Liver Cancer cfDNA sequencing data and established a flexible framework for hypothesis-driven studies of cfDNA fragmentation mechanisms.
- **Manuscript in preparation.**
- Supervisor: Dr. Mahya Mehrmohammadi, Department of Biotechnology, Tehran University

Visiting Research Assistant, Synchrotron-light for Experimental Science and Applications in the Middle East (SESAME) **Allan, Jordan**
BEATS Beamline February 2025

- Worked on-site, using synchrotron-based micro-CT to obtain sub-micron resolution internal imaging of two insect species (*Macrolophus pygmaeus*, and *Nesidiocoris tenuis*)
- Prepared and mounted insect specimens, ensuring optimal orientation, contrast, and beam compatibility
- Supported data acquisition, preliminary reconstruction, and integrity checks of tomographic volumes
- Project Supervisor: Dr Fatemeh Elmi, Department of Marine Physics and Chemistry, University of Mazandaran

Visiting Research Assistant, Soleil Synchrotron **Saint-Aubin, France**
SMIS Beamline February 2024

- Assisted in operating synchrotron infrared microspectroscopy instrumentation for high-resolution biochemical mapping of mouse skin wound sections treated with two collagen-based skin therapies
- Contributed to data collection and spectral quality control for biochemical signature comparison
- Manuscript in process
- Project Supervisor: Dr Fatemeh Elmi, Department of Marine Physics and Chemistry, University of Mazandaran

Visiting Research Assistant, Elettra Sincrotrone Trieste **Trieste, Italy**
BEAR Beamline September 2022

- Assisted in performing X-ray absorption spectroscopy (XAS) and X-ray fluorescence (XRF) to analyze structural and chemical properties of collagen type I intercalated with talc layers
- Worked on capturing intrinsic fluorescence signatures from collagen composites to evaluate nanoscale interactions and structural integrity
- Work contributed to the study published in *Journal of Molecular Structure* (2023).
- Project Supervisor: Dr Fatemeh Elmi, Department of Marine Physics and Chemistry, University of Mazandaran

PUBLICATION

Darbazi M., Elmi F., Elmi M.M, Giglia A., **Hoda A.**, Synchrotron X-ray absorption spectroscopy and fluorescence spectroscopy studies of collagen type I-talc intercalated nanocomposites, Journal of Molecular Structure, Volume 1285, 2023, 135558

Elmi, M.M., Elmi, F., Longo, E., **Hoda, A**., Farnam Taheri, A., Tromba, G., Paiva, K. Synchrotron X-ray Phase-Contrast Microtomography of Dermal Fibrous Matrix Remodeling in Rat Wounds Treated with Chitosan-Based Films. Under review, Academic Radiology.

RELEVANT SKILLS

Computational Biology: Omics/NGS analysis (WES, ATAC-seq, MNase-seq), Stochastic modelling, Bioinformatics, Machine Learning

Genome Engineering: CRISPR Design, Recombinant DNA/Vector Design, Protein Engineering

Experimental Foundations: Cell and Tissue Culture, Nanobiotechnology, Neurophysiology, Genetics, Nucleic Acid Technology, Cellular and Molecular Immunology, Stem Cells and Regenerative Medicine, Precision Medicine

Additional Activities: Neuromatch Academy (Computational Neuroscience, self-study); EHIA Neuroscience Program; CS50x Harvard Puzzle Day Contest 2023; top 0.1% rank in the National Entrance Exam

LANGUAGES

English, French, basic Dutch, basic German, basic Japanese, basic Chinese

REFERENCES

Dr. Mahya Mehrmohammadi — Department of Biotechnology, University of Tehran , Iran
(M.S. Supervisor) — mehrmohamadi@ut.ac.ir

Dr. Sayed-Amir Marashi — Department of Biotechnology, University of Tehran , Iran
(Advisor) — marashi@ut.ac.ir

Dr. Fatemeh Elmi — Department of Marine Chemistry, University of Mazandaran, Iran
(Project Supervisor) — f.elmi@umz.ac.ir